Question	Answer	Marks	Partial Marks
7(a)(ii)	[a =] –3 [b =] 8 [c =] 24	**3**	**FT** *their* **(a)(i)** for a and b **B1FT** for each of a and b or both correct but reversed **B1** for [c =] 24
7(a)(iii)	8	**3**	**M2** for $5 - 2x$ or **M1** for $-2x$ or $5 - kx$, $k \neq 0$
7(b)(i)	Correct sketch: positive cubic shape and max on the y-axis or to the right of y-axis with one root at (–1, 0) and turning point at (3, 0) and y-intercept at (0, 9) all labelled	**4**	**B1** for positive cubic shape with max on the y-axis or to the right of y-axis **B1** for root at (–1, 0) **B1** for turning point at (3, 0) **B1** for y-intercept (0, 9) If 0 score **SC1** for all three intercepts on axes identified
7(b)(ii)	$x^3 - 5x^2 + 3x + 9$ final answer	**3**	**B2** for correct expansion of three brackets unsimplified **B1** for correct expansion of two brackets with at least 3 terms correct
8(a)(i)	$\begin{pmatrix} 4 \\ 4 \end{pmatrix}$	**2**	**B1** for $\begin{pmatrix} 4 \\ k \end{pmatrix}$ or $\begin{pmatrix} k \\ 4 \end{pmatrix}$
8(a)(ii)	$\begin{pmatrix} -4 \\ 8 \end{pmatrix}$	**2**	**B1** for $\begin{pmatrix} -4 \\ k \end{pmatrix}$ or $\begin{pmatrix} k \\ 8 \end{pmatrix}$
8(a)(iii)	5.39 or 5.385..	**2**	**M1** for $(-2)^2 + 5^2$ oe
8(b)(i)	$\mathbf{a} + \mathbf{b}$	**1**	
8(b)(ii)	$\frac{3}{2}\mathbf{a} + \mathbf{b}$	**2**	**M1** for a correct route, e.g. $\overrightarrow{OA} + \overrightarrow{AE}$
8(b)(iii)	$2\mathbf{a} + \frac{4}{3}\mathbf{b}$	**3**	**M2** for unsimplified $\overrightarrow{OD}$ or for $\frac{4}{3}\mathbf{b}$ or **M1** for $\overrightarrow{OD}$ attempted in terms of **a** and **b** or for $\overrightarrow{CD} = \frac{1}{3}\mathbf{b}$ or $\overrightarrow{DB} = \frac{2}{3}\mathbf{b}$ seen
9(a)	2, 3, 4, 5	**2**	**B1** for 3 correct and no extra or 4 correct and one extra or **M1** for $1 < x \leqslant 5$
9(b)(i)	$3y\,(2y - 5x)$	**2**	**B1** for $3(2y^2 - 5xy)$ or $y\,(6y - 15x)$ or for the correct answer seen and then spoiled

Question	Answer	Marks	Partial Marks
9(b)(ii)	$(y - 3x)(y + 3x)$	**2**	**B1** for $(y + 3)(y - 3)$
9(c)	$\frac{4x+5}{(x-1)(2x+1)}$ or $\frac{4x+5}{2x^2-x-1}$ final answer	**3**	**M1** for $3(2x + 1) - 2(x - 1)$ oe isw **M1** for $(x - 1)(2x + 1)$ oe isw
9(d)	(1.74 , 7.21 to 7.24) and (–3.74 , –9.20 to –9.22) cao	**6**	For the y values accept any value rounded to 2 decimal places in the given range **B5** for (1.74 , 7.21 to 7.24) or (–3.74 , –9.20 to –9.22) or $x = 1.74$ and $x = -3.74$ OR **M2** for $2x^2 + 4x - 13 = 0$ or $2y^2 + 4y - 133 = 0$ or **M1** for $2x^2 + 7x - 11 = 3x + 2$ or $y = 2\left(\frac{y-2}{3}\right)^2 + 7\left(\frac{y-2}{3}\right) - 11$ AND **FT** *their* quadratic expression (not $2x^2 + 7x - 11$) **M2FT** for $\frac{-4 \pm \sqrt{4^2 - 4\times2\times-13}}{2\times2}$ or $-1 \pm \sqrt{\frac{15}{2}}$ oe or **M1FT** for $\sqrt{4^2 - 4\times2\times-13}$ oe or for $\frac{-4+\sqrt{k}}{2\times2}$ or $\frac{-4-\sqrt{k}}{2\times2}$ or $(x + 1)^2$ [– 13/2 – 1 = 0]
10(a)	–23	**2**	**M1** for $4 - 3(3^x)$ oe soi
10(b)	$\frac{4-x}{3}$ oe final answer	**2**	**M1** for $x = 4 - 3y$ or $y + 3x = 4$ or $x + 3y = 4$ or $\frac{y}{-3} = \frac{4}{-3} + x$ oe or $\frac{x}{-3} = \frac{4}{-3} + y$ oe
10(c)(i)	$1 + 6x$ final answer	**2**	**M1** for $4 - 3(1 - 2x)$